



Chapter 2: Angles

Three-week revision booklet • 6 worksheets • 108 questions

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal • MAO-WM-01

NAME _____

How to use this booklet

There are six worksheets, two for each week. Each one drills a single skill from the Angles topic test. Each sheet has the same shape: **18 questions**, split into ● **Easy**, ● **Medium** and ● **Challenging**. Work from the top of the page down — the questions get harder as you go. Most questions come with a diagram. The challenging questions give you a box to draw your own diagram before you work — always sketch first. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Naming and classifying angles
1	2	Complementary and supplementary angles
2	3	Angles on a line and at a point
2	4	Vertically opposite angles
3	5	Parallel lines and a transversal
3	6	Mixed revision — all skills

Progress tracker

Worksheet	Date completed	Score / 18	Checked by
1 Naming and classifying angles			
2 Complementary and supplementary angles			
3 Angles on a line and at a point			
4 Vertically opposite angles			
5 Parallel lines and a transversal			
6 Mixed revision — all skills			

Worksheet 1: Naming and classifying angles

Week 1 · Worksheet 1 of 6 · 18 questions

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal · MAO-WM-01

NAME _____

PAGE 2 OF 13

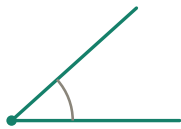
Every angle has a name based on its size. Learn the six types and you can name any angle on sight.

EASY

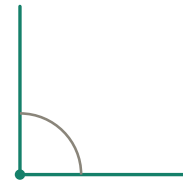
8 QUESTIONS

Name each type of angle.

Hint: compare each angle with a right angle (90°) and a straight angle (180°). Smaller than 90° is **acute**; between 90° and 180° is **obtuse**; bigger than 180° is **reflex**.



1. type = _____



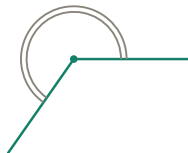
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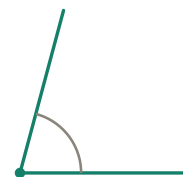
3. type = _____



4. type = _____



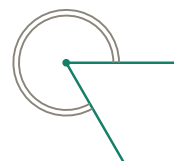
5. type = _____



6. type = _____



7. type = _____



8. type = _____

Worksheet 1: Naming and classifying angles

Week 1 · Worksheet 1 of 6 · 18 questions · continued

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal · MAO-WM-01

NAME _____

PAGE 3 OF 13

MEDIUM

6 QUESTIONS

Name the type of angle for each size.

Hint: you do not need a diagram — just decide where the size sits between 0° , 90° , 180° and 360° .

- | | | |
|----------------------|----------------------|----------------------|
| 1. 47° _____ | 2. 90° _____ | 3. 154° _____ |
| 4. 285° _____ | 5. 180° _____ | 6. 360° _____ |

CHALLENGING

4 QUESTIONS

Answer each question. Draw a rough diagram to help you.

Hint: for the clock, a full turn is 360° and there are 60 minutes in a full turn.

7. Draw and label an obtuse angle of about 120° .

8. An angle is 5° less than a right angle. Draw it, then give its size and type.

9. Draw a reflex angle and mark clearly which angle you mean.

10. Through how many degrees does the minute hand of a clock turn in 20 minutes? Draw a clock face to help you.

Worksheet 2: Complementary and supplementary

Week 1 · Worksheet 2 of 6 · 18 questions

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal · MAO-WM-01

NAME _____

PAGE 4 OF 13

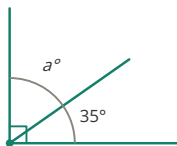
Two angles that add to 90° are **complementary**. Two angles that add to 180° are **supplementary**.

EASY

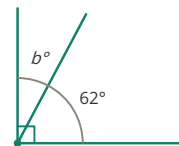
8 QUESTIONS

Find the value of the pronumeral.

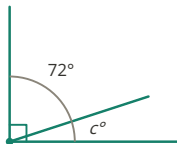
Hint: the small square means the two angles make 90° . A straight line means they make 180° . Subtract to find the unknown.



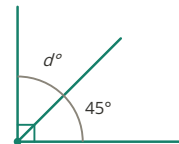
1. $a =$ _____



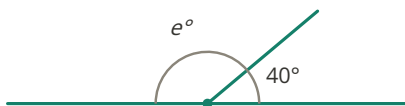
2. $b =$ _____



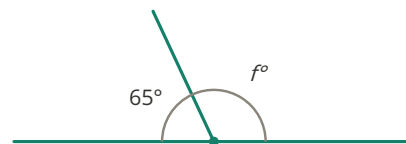
3. $c =$ _____



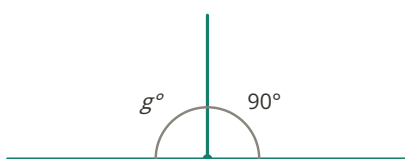
4. $d =$ _____



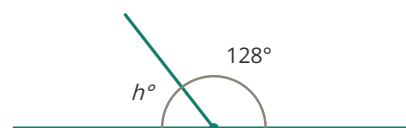
5. $e =$ _____



6. $f =$ _____



7. $g =$ _____



8. $h =$ _____

Worksheet 2: Complementary and supplementary

Week 1 • Worksheet 2 of 6 • 18 questions • continued

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal • MAO-WM-01

NAME _____

PAGE 5 OF 13

MEDIUM

6 QUESTIONS

Answer each question.

Hint: complement means $90 - \text{the angle}$. Supplement means $180 - \text{the angle}$.

1. Write the complement of 25°

2. Write the complement of 68°

3. Write the supplement of 30°

4. Write the supplement of 115°

5. Two complementary angles are equal. Find each one. _____
6. Two supplementary angles are equal. Find each one. _____

CHALLENGING

4 QUESTIONS

Answer each question. Draw a diagram first.

Hint: label the smaller angle x° on your diagram, write the other angle in terms of x , then form an equation that adds to 90° or 180° .

7. An angle is 30° more than its complement. Find both angles.
8. An angle is twice its supplement. Find both angles.

9. The complement of an angle is 25° . Find the supplement of that same angle.
10. Can an obtuse angle have a complement? Draw a diagram and explain.

Worksheet 3: Angles on a line and at a point

Week 2 · Worksheet 3 of 6 · 18 questions

NSW syllabus outcome

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NAME _____

PAGE 6 OF 13

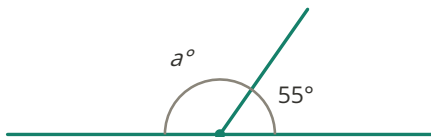
Angles on a straight line add to 180° . Angles that meet at a point add to 360° .

EASY

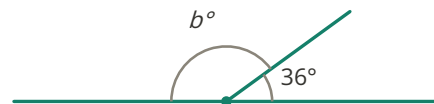
8 QUESTIONS

Find the value of the pronumeral, and give a reason.

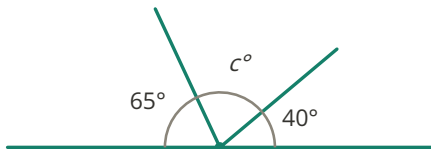
Hint: add up the angles you are given, then subtract from 180° for a straight line or 360° for a point.



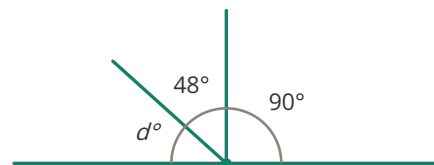
1. $a =$ _____



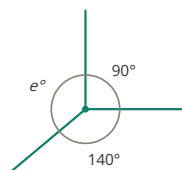
2. $b =$ _____



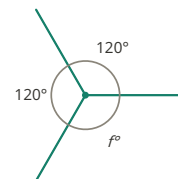
3. $c =$ _____



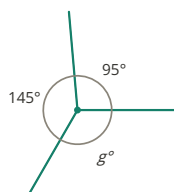
4. $d =$ _____



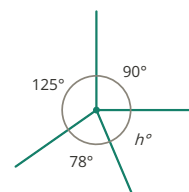
5. $e =$ _____



6. $f =$ _____



7. $g =$ _____



8. $h =$ _____

Worksheet 3: Angles on a line and at a point

Week 2 • Worksheet 3 of 6 • 18 questions • continued

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal • MAO-WM-01

NAME _____

PAGE 7 OF 13

MEDIUM

6 QUESTIONS

Answer each question.

Hint: for the last two, add the expressions together and set the total equal to 180° or 360° , then solve for x .

- Three angles on a straight line are a° , 65° and 48° . Find a . _____
- Four angles at a point are 90° , 125° , 78° and m° . Find m . _____
- Two angles on a straight line are equal. Find each one. _____
- Three equal angles meet at a point. Find each one. _____
- Angles on a line are $(3x)^\circ$ and $(2x)^\circ$. Find x . _____
- Angles at a point are $(4x)^\circ$, $(5x)^\circ$ and 270° . Find x . _____

CHALLENGING

4 QUESTIONS

Answer each question. Draw and label a diagram first.

Hint: sketch the straight line or the point first, then write the angles on it before you form an equation.

- Two angles on a straight line are in the ratio 1 : 2. Find both angles.
- Two angles on a straight line differ by 40° . Find both angles.

- Angles at a point are x° , $2x^\circ$ and $3x^\circ$. Find all three.

- Explain why two right angles on a straight line leave no angle over.

Worksheet 4: Vertically opposite angles

Week 2 · Worksheet 4 of 6 · 18 questions

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal · MAO-WM-01

NAME _____

PAGE 8 OF 13

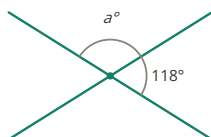
When two straight lines cross, the angles opposite each other are **equal**. Angles next to each other still add to 180° .

EASY

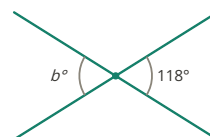
8 QUESTIONS

Find the value of the pronumeral, and give a reason.

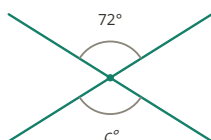
Hint: first decide whether the two angles are **opposite** each other (equal) or **next to** each other on a line (add to 180°).



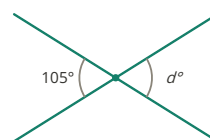
1. $a =$ _____



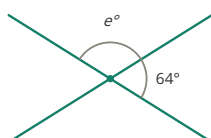
2. $b =$ _____



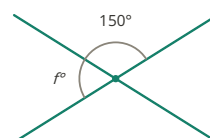
3. $c =$ _____



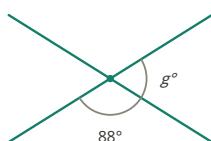
4. $d =$ _____



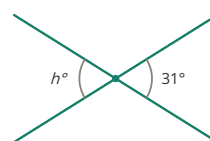
5. $e =$ _____



6. $f =$ _____



7. $g =$ _____



8. $h =$ _____

Worksheet 4: Vertically opposite angles

Week 2 • Worksheet 4 of 6 • 18 questions • continued

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal • MAO-WM-01

NAME _____

PAGE 9 OF 13

MEDIUM

6 QUESTIONS

Two straight lines cross. Answer each question.

Hint: in a crossing, opposite angles are equal, so there are really only two different sizes.

1. One angle is 68° . Find the other three angles. _____
2. All four angles are equal. Find each one. _____
3. Vertically opposite angles are $(3x)^\circ$ and 75° . Find x .

4. Two angles next to each other are $(5x)^\circ$ and $(4x)^\circ$. Find both angles. _____
5. One angle is a right angle. What are the other three?

6. Explain why vertically opposite angles must always be equal. _____

CHALLENGING

4 QUESTIONS

Answer each question. Draw and label a diagram first.

Hint: if two angles are vertically opposite, set the two expressions equal to each other and solve.

7. Vertically opposite angles are $(2x + 10)^\circ$ and $(3x - 20)^\circ$. Find x .
8. Two lines cross so that one angle is three times another. Find all four angles.
9. Three lines meet at one point, making six angles. If one is 40° , name an angle that must also be 40° .
10. Two lines cross. One angle is 47° . Find the other three, giving a reason for each.

Worksheet 5: Parallel lines and a transversal

Week 3 • Worksheet 5 of 6 • 18 questions

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal • MAO-WM-01

NAME _____

PAGE 10 OF 13

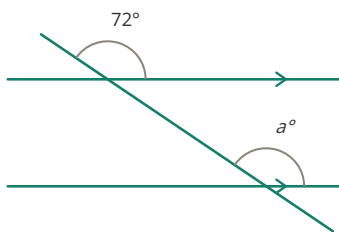
On parallel lines, **corresponding** angles are equal, **alternate** angles are equal, and **co-interior** angles add to 180° . The arrows show the lines are parallel.

EASY

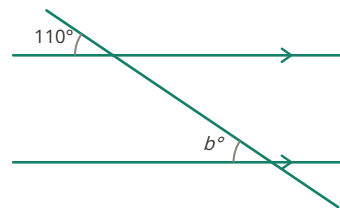
8 QUESTIONS

Find the value of the pronumeral and name the angle relationship used.

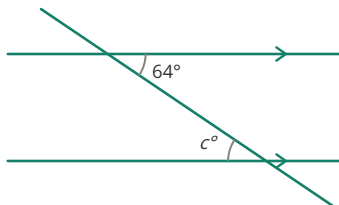
Hint: look for the shape the two angles make. An **F** shape means corresponding (equal), a **Z** shape means alternate (equal), and a **C** or **U** shape means co-interior (add to 180°).



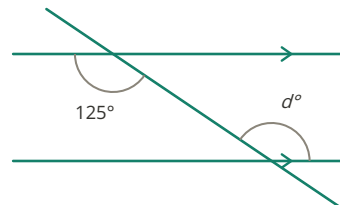
1. $a =$ _____



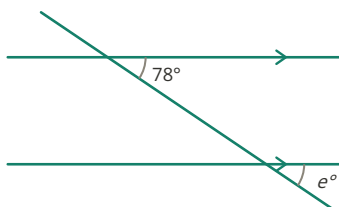
2. $b =$ _____



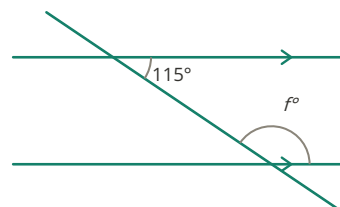
3. $c =$ _____



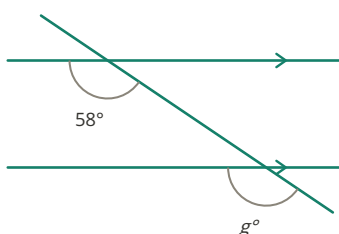
4. $d =$ _____



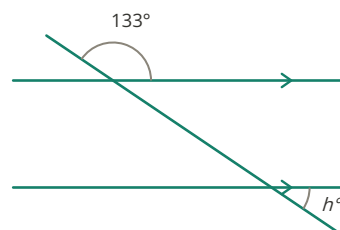
5. $e =$ _____



6. $f =$ _____



7. $g =$ _____



8. $h =$ _____

Worksheet 5: Parallel lines and a transversal

Week 3 · Worksheet 5 of 6 · 18 questions · continued

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal · MAO-WM-01

NAME _____

PAGE 11 OF 13

MEDIUM

6 QUESTIONS

The lines are parallel. Answer each question.

Hint: only co-interior angles need subtracting from 180° . Corresponding and alternate angles are simply equal.

- Corresponding angles are 118° and x° . Find x .

- Alternate angles are 47° and y° . Find y . _____
- Co-interior angles are 62° and z° . Find z . _____
- Two co-interior angles are equal. Find each one.

- Corresponding angles are $(4x)^\circ$ and 100° . Find x .

- Explain the difference between alternate and co-interior angles. _____

CHALLENGING

4 QUESTIONS

Answer each question. Draw and label a diagram first.

Hint: draw two parallel lines with arrows on them, then draw the transversal across both before marking any angles.

- Two co-interior angles between parallel lines are x° and 118° . Find x .
- A transversal makes a 65° angle with one of two parallel lines. Find all eight angles formed.

- Co-interior angles are $(3x)^\circ$ and $(2x)^\circ$. Find x and both angles.
- A transversal crosses two lines making corresponding angles of 70° and 75° . Are the lines parallel? Explain.

Worksheet 6: Mixed revision

Week 3 · Worksheet 6 of 6 · 18 questions

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal · MAO-WM-01

NAME _____

PAGE 12 OF 13

This sheet mixes every skill from the Angles chapter. Work through it as though it were the topic test.

EASY

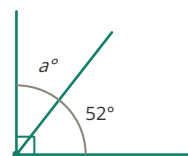
8 QUESTIONS

Find the value of the pronumeral, and give a reason.

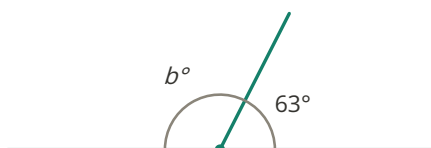
Hint: for each diagram, first decide which rule applies — straight line, point, vertically opposite, or parallel lines.



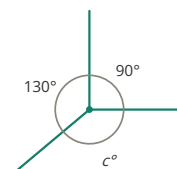
1. type = _____



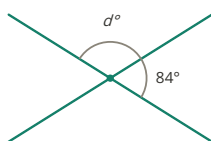
2. $a =$ _____



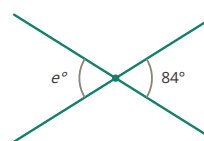
3. $b =$ _____



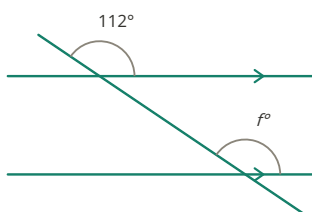
4. $c =$ _____



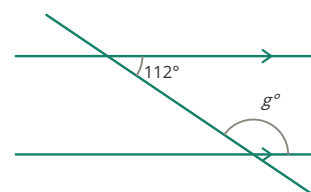
5. $d =$ _____



6. $e =$ _____



7. $f =$ _____



8. $g =$ _____

Worksheet 6: Mixed revision

Week 3 · Worksheet 6 of 6 · 18 questions · continued

NSW syllabus outcome

MA4-ANG-C-01 — identifies and uses angle relationships, including those on parallel lines cut by a transversal · MAO-WM-01

NAME _____

PAGE 13 OF 13

● MEDIUM

6 QUESTIONS

Answer each question.

Hint: a quick sketch makes each of these much easier — mark the angle you know first, then work around.

- Write the complement of 41° . _____
- Write the supplement of 128° . _____
- Three angles on a straight line are a° , 65° and 48° . Find a . _____
- Four angles at a point are 90° , 125° , 78° and m° . Find m . _____
- Two co-interior angles between parallel lines are x° and 118° . Find x . _____
- Two lines cross. One angle is 47° . Find the other three. _____

● CHALLENGING

4 QUESTIONS

Answer each question. Draw and label a diagram first.

Hint: for the last question, think about what two acute angles would add to, and compare that with 180° .

- An angle is three times its complement. Find both angles.
- Through how many degrees does the minute hand of a clock turn in 25 minutes?

- Angles on a straight line are $(3x)^\circ$, $(2x)^\circ$ and 30° . Find x .
- Explain why two co-interior angles between parallel lines can never both be acute.